

Limits (極限)

極根的概念 Idea of Limit

$\lim_{x\to a} f(x)$ 代表當 x 接近 (close to) a 時 (符號以 $x \to a$ 表示) $f(x)$ 的數值。

e.g. 若 $x \to 2,\ x+3 \to 5$.所以 $\lim_{x\to 2} (x+3) = 5$.

若 $x \to -3,\ x^2 \to 9$. 所以 $\lim_{x\to -3} x^2 = 9$.

若 $x \to 0,\ \frac{1}{x^2} \to \infty$. 所以 $\lim_{x\to 0} \frac{1}{x^2} = \infty$.

極根的重要性 Importance of Limit

試考慮 $f(x) = \frac{x^2-4}{x-2}$

x	1.9	1.99	1.999	2	2.001	2.01	2.1
$f(x)$	3.9	3.99	3.999	undefined	4.001	4.01	4.1

當 x 等於 **(equal to)** 2 時，$f(x)$ 是未下定義 (undefined) $f(2)$ is undefined

當 x 接近 **(close to)** 2 時， $f(x)$ 的值接近 4 $\lim_{x\to 2} f(x) = 4$

雖然 $f(2)$ 未下定義 …

但使用極限，我們知道當 x 接近 2 時， $f(x)$ 會接近 4, i.e. $\lim_{x\to 2} f(x) = 4$

所以運用極限，我們便可以知道一些函數的數值，尤其是當它們是不能定義的時候！



Finding Limits (I)

Method 1: Direct Substitution 直接代入法

我們可以使用以下定義去求出極限 :

$$\lim_{x \to a} f(x) = f(a)$$

e.g. $\lim_{x \to 2} (5x + 3) = 5(2) + 3 = 13$

$\lim_{x \to 1} (x^2 + 5x - 3) = 1 + 5 - 3 = 3$

$\lim_{x \to 3} \dfrac{x-2}{x+2} = \dfrac{3-2}{3+2} = \dfrac{1}{5}$

但是使用直接代入法前，我們或許需要化簡算式 :

e.g. Evaluate 計算 $\lim_{x \to 0} \dfrac{2x^2 + 2x}{x}$

$$\lim_{x \to 0} \frac{2x^2 + 2x}{x} = \lim_{x \to 0} \frac{x(2x+2)}{x}$$
$$= \lim_{x \to 0} (2x + 2)$$
$$= 2(0) + 2$$
$$= 2$$

我們可以約簡位於分子及分母中的 x (注意 x 不是等於零 !)

$x \to 0$ 代表 x 是接近零，而不是等於零 !

所以 $\lim_{x \to 0} \dfrac{2x^2 + 2x}{x}$ 理論上是能夠計算的。我們可以約簡位於分子及分母中的 x ，然後去計算極限。



Finding limits (I)



Evaluate the following limits :

1. $\lim_{x\to 1} (x+1) =$

2. $\lim_{x\to 1} (x-1) =$

3. $\lim_{x\to 1} (k-1) =$

4. $\lim_{x\to 0} 3 =$

5. $\lim_{x\to 0} \frac{x}{x} =$

6. $\lim_{x\to 0} \frac{x^2}{x} =$

7. $\lim_{x\to 0} \frac{x}{x^3} =$

8. $\lim_{x\to 0} \frac{x^3+x}{x} =$

9. $\lim_{x\to 0} \frac{x}{x^3+x} =$

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Finding Limits (2)

Method 2: Factorization 因式分解

若果用直接代入法會使算式不能計算 (undefined)，我們或許需要使用因式分解 (factorization) 去才用 $\lim_{x\to a} f(x) = f(a)$ 去計算極限。

e.g. Evaluate 計算 $\lim_{x\to 2} \dfrac{x^2-4}{x-2}$.

錯誤方法 ***Wrong approach***

$$\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} \frac{(2)^2-4}{2-2}$$

$$= \lim_{x\to 2} \frac{0}{0}$$

✗

正確方法 ***Good approach***

$$\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} \frac{(x+2)(x-2)}{x-2}$$

$$= \lim_{x\to 2} (x+2)$$

$$= 2+2$$

$$= 4$$

✓



Note: 若你發覺代入法會使分母變成零的話，便請多使用因式分解！

Useful identities for factorization 因式分解時常用的恒等式：

1. $a^2-b^2=(a+b)(a-b)$

2. $a^3+b^3=(a+b)(a^2-ab+b^2)$

3. $a^3-b^3=(a-b)(a^2+ab+b^2)$

Finding Limits (2)

Let's TRY IT! Evaluate the following limits :

1. $\lim\limits_{x\to 2} \dfrac{x^3-1}{x-1} =$

2. $\lim\limits_{x\to 1} \dfrac{x^3-1}{x-1} =$

3. $\lim\limits_{x\to -2} \dfrac{x+2}{x^3+8} =$

4. $\lim\limits_{x\to 4} \dfrac{x^2-x-12}{x^2-16} =$

5. $\lim\limits_{x\to 9} \dfrac{\sqrt{x}-3}{x-9} =$

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Finding Limits (3)

Method 3: Multiplying conjugate (for surd form)

乘以共軛（根式專用）

e.g. Evaluate $\lim\limits_{x\to 0}\dfrac{\sqrt{1+x}-1}{x}$.

$$\lim_{x\to 0}\frac{\sqrt{1+x}-1}{x} = \lim_{x\to 0}\frac{\sqrt{1+x}-1}{x}\cdot\frac{\sqrt{1+x}+1}{\sqrt{1+x}+1}$$

$$= \lim_{x\to 0}\frac{(\sqrt{1+x})^2-1^2}{x(\sqrt{1+x}+1)}$$

$$= \lim_{x\to 0}\frac{1+x-1}{x(\sqrt{1+x}+1)}$$

$$= \lim_{x\to 0}\frac{x}{x(\sqrt{1+x}+1)}$$

$$= \lim_{x\to 0}\frac{1}{\sqrt{1+x}+1}$$

$$= \frac{1}{\sqrt{1+0}+1}$$

$$= \frac{1}{2}$$

$\sqrt{a}+b$ 的共軛 (conjugate) 是 $\sqrt{a}-b$

$(\sqrt{a}+b)(\sqrt{a}-b)$

$= a-b^2$

Check:

分母現在不是等於零，

故可用代入法

More about conjugate 更多共軛的知識:

除了求出極限外，共軛在分數有理化 (rationalization) 的運算上亦是非常有用的:

e.g. $\dfrac{4}{2-\sqrt{3}} = \dfrac{4}{2-\sqrt{3}}\cdot\dfrac{2+\sqrt{3}}{2+\sqrt{3}} = \dfrac{4(2+\sqrt{3})}{4-3} = 8+4\sqrt{3}$

Finding Limits (3)



Evaluate the following limits :

1. $\lim\limits_{x\to 9} \dfrac{\sqrt{x}-2}{x-4} =$

2. $\lim\limits_{x\to 4} \dfrac{\sqrt{x}-2}{x-4} =$

3. $\lim\limits_{x\to 1} \dfrac{x-1}{\sqrt{x^2+3}-2} =$

4. $\lim\limits_{h\to 0} \dfrac{\sqrt{x+h}-\sqrt{x}}{h} =$

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Limits at Infinity (無限)

當 x 接近無限 **(x tends to infinity)** 時，以下定理可以幫到你 :

$$\lim_{x\to\infty}\frac{1}{x}=0$$

(同理 $\lim_{x\to-\infty}\frac{1}{x}=0$)

e.g.1 $\lim_{x\to\infty}\frac{1}{x^n}=0$ 其中 n 是正整數

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e.g.2
$$\lim_{x\to\infty}\frac{x}{x+2}=\lim_{x\to\infty}\left(\frac{x}{x+2}\cdot\frac{\frac{1}{x}}{\frac{1}{x}}\right)$$
$$=\lim_{x\to\infty}\frac{1}{1+\frac{2}{x}}$$
$$=\frac{1}{1+0}$$
$$=1$$

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e.g.3
$$\lim_{x\to\infty}\frac{x+1}{x^2+2x+3}=\lim_{x\to\infty}\left(\frac{x+1}{x^2+2x+3}\cdot\frac{\frac{1}{x^2}}{\frac{1}{x^2}}\right)$$
$$=\lim_{x\to\infty}\frac{\frac{1}{x}+\frac{1}{x^2}}{1+\frac{2}{x}+\frac{3}{x^2}}$$
$$=\frac{0+0}{1+0+0}$$
$$=0$$



Techniques:

嘗試將分子與分母同時乘以 $\frac{1}{x^n}$，其中 n 是算式中最大的指數。跟著就可以運用 $\lim_{x\to\infty}\frac{1}{x^n}=0$ 去計算極限的值了。

Limits at Infinity



Evaluate the following limits :

1. $\lim_{x\to\infty} (x+1) =$

2. $\lim_{x\to\infty} (\frac{1}{x}+1) =$

3. $\lim_{x\to\infty} \frac{x+1}{x} =$

4. $\lim_{x\to\infty} \frac{x^2+x+1}{2x^2} =$

5. $\lim_{x\to\infty} \frac{x-1}{x^2-1} =$

6. $\lim_{x\to\infty} \frac{x^2-1}{x-1} =$

7. $\lim_{x\to\infty} \frac{x-1}{x^2-3x+2} =$

8. $\lim_{x\to\infty} \frac{2x^2+4x-1}{x^2-3x+2} =$

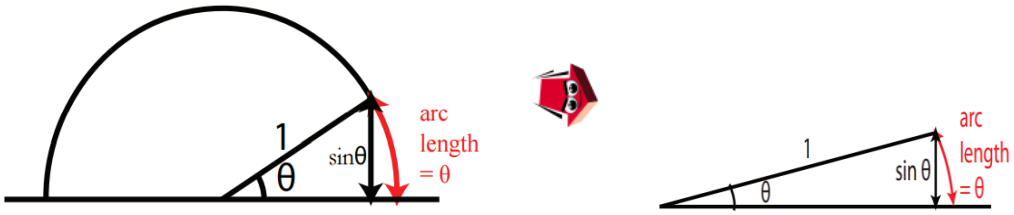
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Limits related to Sine Function (正弦之極限)

還記得若某扇形 (Sector) 的其弧長計算方式如下

$$\text{arc length} = 2\pi r \times \frac{\theta}{2\pi} = r\theta \text{ ， 其中圓心角 } \theta \text{ 是以弧度 (radian) 計算}$$

現在若我們考慮一個半徑為 1 單位的圓，可見到當圓心角 θ 的值接近零時，直角三角形的對邊值 ($\sin\theta$) 與弧長值 (θ) 會十分接近。



所以我們便有以下關於正弦 (sine) 的極限：

$$\lim_{\theta\to 0}\frac{\sin\theta}{\theta} = 1$$

(或 $\lim_{x\to 0}\frac{\sin x}{x} = 1$)

e.g.1 $\lim_{x\to 0}\frac{\sin 3x}{2x} = \lim_{x\to 0}\left(\frac{\sin 3x}{3x}\cdot\frac{3}{2}\right)$

$= \frac{3}{2}\lim_{x\to 0}\frac{\sin 3x}{3x}$

$= \frac{3}{2}$

e.g.2 $\lim_{x\to 0}\frac{\tan 3x}{\sin 4x} = \lim_{x\to 0}\left(\frac{\sin 3x}{\cos 3x}\cdot\frac{1}{\sin 4x}\right)$

$= \lim_{x\to 0}\left(\frac{\sin 3x}{3x}\cdot\frac{4x}{\sin 4x}\cdot\frac{3x}{4x}\cdot\frac{1}{\cos 3x}\right)$

$= (\lim_{x\to 0}\frac{3}{4})\cdot(\lim_{x\to 0}\frac{\sin 3x}{3x})\cdot(\lim_{x\to 0}\frac{4x}{\sin 4x})\cdot(\lim_{x\to 0}\frac{1}{\cos 3x})$

$= \frac{3}{4}$

Techniques:

嘗試將算式改寫為 $\frac{\sin nx}{nx}$ 或 $\frac{nx}{\sin nx}$ 的形式，跟著就可以運用 $\lim_{x\to 0}\frac{\sin\theta}{\theta} = 1$ 去計算極限的值了。

Finding Limits (3)

Let's TRY IT! Evaluate the following limits :

1. $\lim\limits_{x\to 0} \dfrac{\sin 3x}{x} =$

2. $\lim\limits_{x\to 0} \dfrac{\tan 2x}{x} =$

3. $\lim\limits_{x\to 0} \dfrac{x\tan x}{\sin^2 2x} =$

4. $\lim\limits_{x\to 0} \dfrac{1-\cos 2x}{x^2} =$

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Theorems on limits

當進行極限運算時，你可使用以下定理 :

1. $\lim\limits_{x\to a} k = k$ if k is constant 常數

2. $\lim\limits_{x\to a} [k \cdot f(x)] = k \cdot \lim\limits_{x\to a} f(x)$ if k is constant 常數

3. $\lim\limits_{x\to a} \left[f(x) \pm g(x)\right] = \lim\limits_{x\to a} f(x) \pm \lim\limits_{x\to a} g(x)$

4. $\lim\limits_{x\to a} \left[f(x) \cdot g(x)\right] = \lim\limits_{x\to a} f(x) \cdot \lim\limits_{x\to a} g(x)$

5. $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim\limits_{x\to a} f(x)}{\lim\limits_{x\to a} g(x)}$ (given 已知 $\lim\limits_{x\to a} g(x) \neq 0$)

6. $\lim\limits_{x\to a} \sqrt{f(x)} = \sqrt{\lim\limits_{x\to a} f(x)}$

7. $\lim\limits_{x\to \infty} \dfrac{1}{x} = 0$

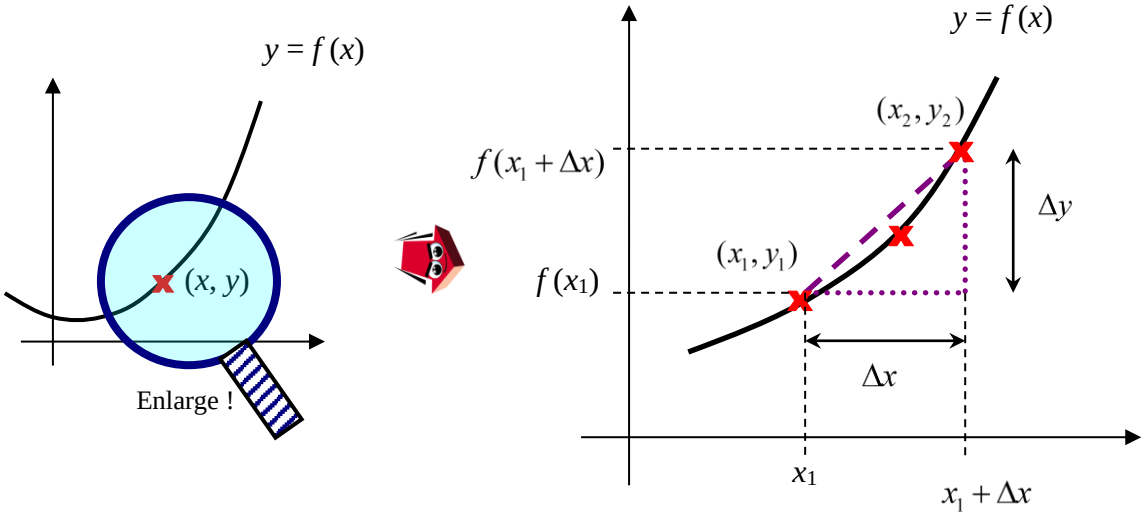
8. $\lim\limits_{x\to 0} \dfrac{\sin x}{x} = 1$

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First principle of Differentiation (微分基本原理)

Idea of Differentiation 微分的概念

Q : Can we find the slope of a given point on the curve?
我們可不可以求出曲線上某一點的斜率？



我們知道 $\text{斜率} = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{\Delta y}{\Delta x}$ (Δx 代表 x 值之差， Δy 的意義亦相同)

我們要求曲線上某一點 (x, y) 的斜率的話， 我們需要找任何兩點 (x_1, y_1) 及 (x_2, y_2) ，它們均需非常接近 (x, y)。因為兩點 (x_1, y_1) 及 (x_2, y_2) 非常接近，所以 $x_2 - x_1 \to 0$， i.e. $\Delta x \to 0$ (同理 $\Delta y \to 0$).

利用極限的概念，我們得知 $\text{斜率 (Slope)} = \lim\limits_{\Delta x \to 0} \dfrac{\Delta y}{\Delta x}$.

從圖中得知 $\Delta y = f(x_1 + \Delta x) - f(x_1)$ ，因為 $x_1 \approx x$ ，所以我們定義以下微分：

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

這就是微分基本原理 (First Principle of Differentiation)。

First principle of Differentiation

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

e.g.1 If $y = 2x^2 + 1$, find $\frac{dy}{dx}$ from first principles.

$$\begin{aligned}
\frac{dy}{dx} &= \lim_{\Delta x \to 0} \frac{[2(x+\Delta x)^2 + 1] - [2x^2 + 1]}{\Delta x} \\
&= \lim_{\Delta x \to 0} \frac{2(x^2 + 2x\Delta x + \Delta x^2) - 2x^2}{\Delta x} \\
&= \lim_{\Delta x \to 0} \frac{\Delta x(4x + 2\Delta x)}{\Delta x} \\
&= \lim_{\Delta x \to 0} (4x + 2\Delta x) \\
&= 4x + 2(0) \\
&= 4x
\end{aligned}$$

Techniques:

1. ***Simplify*** 化簡算式
2. ***Cancel*** Δx 約簡 Δx
3. ***Substitution*** 直接代入

e.g.2 If $y = \frac{1}{2x-1}$, find $\frac{dy}{dx}$ from first principles.

$$\begin{aligned}
\frac{dy}{dx} &= \lim_{\Delta x \to 0} \frac{1}{\Delta x} \left(\frac{1}{2(x+\Delta x) - 1} - \frac{1}{2x-1} \right) \\
&= \lim_{\Delta x \to 0} \frac{1}{\Delta x} \left(\frac{(2x-1) - [2(x+\Delta x) - 1]}{[2(x+\Delta x) - 1]\,(2x-1)} \right) \\
&= \lim_{\Delta x \to 0} \frac{1}{\Delta x} \cdot \frac{-2\Delta x}{[2(x+\Delta x) - 1](2x-1)} \\
&= \lim_{\Delta x \to 0} \frac{-2}{[2(x+\Delta x) - 1](2x-1)} \\
&= \frac{-2}{[2(x+0) - 1](2x-1)} \\
&= \frac{-2}{(2x-1)^2}
\end{aligned}$$

將微分基本原理算式中的分母以 $\frac{1}{\Delta x}$ 表示，那就可避免算式混亂。

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first principle of Differentiation



[A.Math 2007 Q.4]

Find $\frac{d}{dx}(x^2+1)$ from first principles. ***(4 marks)***

[A.Math 2005 Q.3]

Find $\frac{d}{dx}(\frac{1}{x})$ from first principles. ***(4 marks)***



First principle of Differentiation

e.g.3 If $y=\sqrt{x^2+1}$, find $\dfrac{dy}{dx}$ from first principles.

$$\frac{dy}{dx}=\lim_{\Delta x\to 0}\frac{\sqrt{(x+\Delta x)^2+1}-\sqrt{x^2+1}}{\Delta x}$$

$$=\lim_{\Delta x\to 0}\frac{\sqrt{(x+\Delta x)^2+1}-\sqrt{x^2+1}}{\Delta x}\cdot\frac{\sqrt{(x+\Delta x)^2+1}+\sqrt{x^2+1}}{\sqrt{(x+\Delta x)^2+1}+\sqrt{x^2+1}}$$

$$=\lim_{\Delta x\to 0}\frac{[(x+\Delta x)^2+1]\ -\ [x^2+1]}{\Delta x(\sqrt{(x+\Delta x)^2+1}+\sqrt{x^2+1})}$$

$$=\lim_{\Delta x\to 0}\frac{x^2+2x\Delta x+(\Delta x)^2+1-x^2-1}{\Delta x(\sqrt{(x+\Delta x)^2+1}+\sqrt{x^2+1})}$$

$$=\lim_{\Delta x\to 0}\frac{2x\Delta x+(\Delta x)^2}{\Delta x(\sqrt{(x+\Delta x)^2+1}+\sqrt{x^2+1})}$$

$$=\lim_{\Delta x\to 0}\frac{\Delta x(2x+\Delta x)}{\Delta x(\sqrt{(x+\Delta x)^2+1}+\sqrt{x^2+1})}$$

$$=\lim_{\Delta x\to 0}\frac{2x+\Delta x}{\sqrt{(x+\Delta x)^2+1}+\sqrt{x^2+1}}$$

$$=\frac{2x+0}{\sqrt{(x+0)^2+1}+\sqrt{x^2+1}}$$

$$=\frac{2x}{2\sqrt{x^2+1}}$$

$$=\frac{x}{\sqrt{x^2+1}}$$

若算式中一開始便有根式的話，可嘗試先將分子及分母同時乘以共軛。

first principle of Differentiation



[M2 Sample Paper Q.1]

Find $\frac{d}{dx}(\sqrt{2x})$ from first principles. ***(4 marks)***



First principle of Differentiation

e.g.4 If $y = \sin x$, find $\frac{dy}{dx}$ from first principles.

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\sin(x+\Delta x) - \sin x}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{2\cos\frac{(x+\Delta x)+x}{2}\sin\frac{(x+\Delta x)-x}{2}}{\Delta x}$$

$$= 2 \cdot \left[\lim_{\Delta x \to 0} \cos(x+\frac{\Delta x}{2})\right] \cdot \left[\lim_{\Delta x \to 0} \frac{\sin\frac{\Delta x}{2}}{\Delta x}\right]$$

$$= 2 \cdot \left[\lim_{\Delta x \to 0} \cos(x+\frac{\Delta x}{2})\right] \cdot \left[\lim_{\Delta x \to 0} \frac{\sin\frac{\Delta x}{2}}{\frac{\Delta x}{2}} \cdot \frac{1}{2}\right]$$

$$= 2 \cdot \frac{1}{2} \cdot \left[\lim_{\Delta x \to 0} \cos(x+\frac{\Delta x}{2})\right] \cdot \left[\lim_{\Delta x \to 0} \frac{\sin\frac{\Delta x}{2}}{\frac{\Delta x}{2}}\right]$$

$$= \cos(x+\frac{0}{2}) \cdot 1$$

$$= \cos x$$

若需要由微分基本原理微分 $\sin x$ 或 $\cos x$，請善用三角函數和積互化公式 (sum to product formula):

$$\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$$

$$\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$$

first principle of Differentiation



[M2 2013 Q.1]

Find $\frac{d}{dx}(\sin 2x)$ from first principles. ***(4 marks)***

HKDSE Questions (level 1)

1. ***[A.Math 2003 Q.2]***

Find $\frac{d}{dx}(x^3)$ from first principles. ***(4 marks)***

2. ***[M2 2015 Q.1]***

Find $\frac{d}{dx}(x^5+4)$ from first principles ***(4 marks)***

HKCEE Questions (level 1)

3. ***[A.Math 1991 Paper I Q.2]***

Let f(x) = $\frac{1}{1+x}$. Find f'(x) from first principle. ***(5 marks)***

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HKCEE Questions (level 2)

4. ***[A.Math 1988 Paper I Q.1]***

(a) Simplify $(\sqrt{x+1+\Delta x}-\sqrt{x+1})(\sqrt{x+1+\Delta x}+\sqrt{x+1})$

(b) Let $y=\sqrt{x+1}$. Find $\frac{dy}{dx}$ from first principles. ***(5 marks)***

HKCEE Questions (level 2)

5. ***[A.Math 2000 Paper I Q.3]***

(a) Show that $\frac{1}{\sqrt{x+\Delta x}}-\frac{1}{\sqrt{x}}=\frac{-\Delta x}{\sqrt{x}(\sqrt{x+\Delta x})(\sqrt{x}+\sqrt{x+\Delta x})}$.

(b) Find $\frac{d}{dx}(\frac{1}{\sqrt{x}})$ from first principles. ***(5 marks)***

HKDSE Questions (level 2)

6. ***[M2 2016 Q.2]***

Prove that $\frac{1}{\sqrt{x}}-\frac{1}{\sqrt{x+h}}=\frac{h}{(x+h)\sqrt{x}+x\sqrt{x+h}}$. Hence, find $\frac{d}{dx}\sqrt{\frac{3}{x}}$ from first principles. ***(5 marks)***

HKDSE Questions (level 2)

7. ***[M2 2019 Q.2]***

Let $f(x)=\dfrac{10x}{7+3x^2}$. Prove that $f(1+h)-f(1)=\dfrac{4h-3h^2}{10+6h+3h^2}$.

Hence, find $f'(1)$ from first principles. ***(4 marks)***

HKDSE Questions (level 2)

8. ***[M2 2020 Q.2]***

Define $f(x) = \dfrac{x}{\sqrt{2+x}}$ for all $x > -2$. Find $f'(2)$ from first principles. ***(4 marks)***

HKDSE Questions (level 3)

9. ***[M2 2017 Q.2]***

Find $\frac{d}{dx}\sec 6\theta$ from first principles. ***(5 marks)***